

Hackathon Problem Statements

Regd No: 25091A3229, 25091A3212

Problem Statement No: 24

Vaccine Distribution Optimizer Develop a cold-chain vaccine distribution planner using network flow algorithms to optimize the movement of temperature-sensitive vaccines from central storage to remote health centers with minimal wastage. Combine this with demand prediction models based on population and historical uptake data. Use OOP to represent storage facilities, transport vehicles, and vaccine batches with temperature-tracking attributes. This directly supports large-scale immunization drives in geographically challenging rural terrains.

Regd No: 25091A3261, 25091A32A3

Problem Statement No: 27

Doctor-Patient Appointment Scheduling System Design an appointment scheduling system using interval scheduling algorithms to maximize doctor utilization while minimizing patient wait times across multiple specializations and time slots. Apply basic forecasting to predict no-show probability based on historical patient behavior, allowing smart overbooking. Model doctors, patients, and slots using OOP with conflict-resolution logic. This improves efficiency in busy government hospital OPDs where manual scheduling causes significant bottlenecks.

Regd No: 25091A3255, 25091A3258

Problem Statement No: 72

Carbon Footprint Calculator App Design a personal/organizational carbon footprint calculator using an OOP-based activity classification system (transport, electricity, diet, waste) where each activity type computes emissions using standard conversion factors. Aggregate and visualize footprint trends over time to encourage behavior change. This promotes environmental awareness and can be extended for corporate sustainability reporting.

Regd No: 25091A32E5, 25091A32E6

Problem Statement No: 78

Electric Vehicle Charging Station Placement Optimizer Create a facility-location optimization tool using algorithms to determine optimal EV charging station placement based on traffic density, existing station coverage gaps, and population data, maximizing accessibility while minimizing redundant coverage. Use OOP to model stations, coverage zones, and demand points. This supports the growing need for EV infrastructure planning in expanding Indian cities.

Regd No: 25091A3205, 25091A3236

Problem Statement No: 89

Green Building Certification Scorer Create a green building scoring tool using an OOP-based rule/criteria engine that evaluates buildings against sustainability parameters (energy efficiency, water conservation, materials, waste management) to generate a certification-readiness score. Use weighted scoring logic similar to real certification standards like GRIHA/IGBC. This helps architects and builders self-assess sustainability compliance before formal certification processes.

Regd No: 25091A32D7, 25091A32K8

Problem Statement No: 95

Credit Score Simulation Engine Build a credit score simulation tool using a weighted, rule-based scoring engine (implemented via OOP strategy pattern) that estimates creditworthiness based on income, repayment history, and existing debt, mirroring how real credit bureaus operate. Allow users to simulate how financial decisions would affect their score. This promotes financial literacy about credit systems, especially for first-time credit seekers.